Method-6

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* Kleen's Theorem (Part-1)

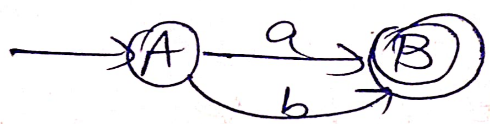
A language is said to be regular, if it is represented by FA.

* Theorem:

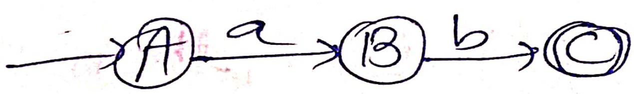
For any regular expression R that represent language L(r), there is a FA that accepts same language.

Proof:

(1) RE : a+b



(2) RE: a.b



(3) RE: $(a/b)^*$

→ To understand Kleen's Theorem, let's take basic defination of RE, where we observe that and a single i/p "a" can be included in RE and corresponding operations that can be performed by the combination as:

→ Let's say 'a' and 'b' be two RE, then

(1) + is a RE to whose corresponding language is $L(a) \cup L(b)$.

(2) • is a RE to whose corresponding language is $L(a) \cdot L(b)$

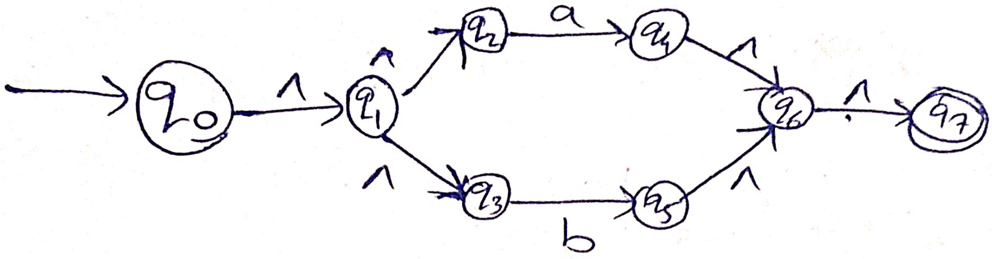
(3) * is a RE to whose corresponding language is $L(a)^* / L(b)^*$

→ Let, S accepts L(a) and T accepts L(b), then R can be represented by combination of S and T as follows:

[Start with Λ, In between put Λ, End with Λ]

(1) $R = S + T$.

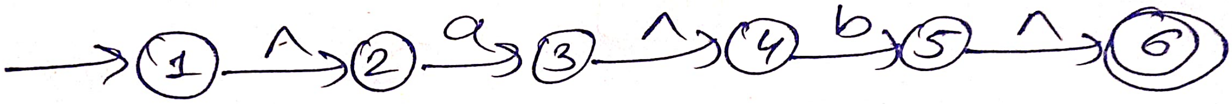
∴ $RE = a + b$. [∵ $S \in L(a)$ & $T \in L(b)$]



[whenever you see, +, / ⇒ Make two paths]

(2) $R = S \cdot T$

∴ $RE = a \cdot b$



(3) $R = S^+ / T^+$

$RE = a^+$ [Make loop of a]

(4) $R = S^*$

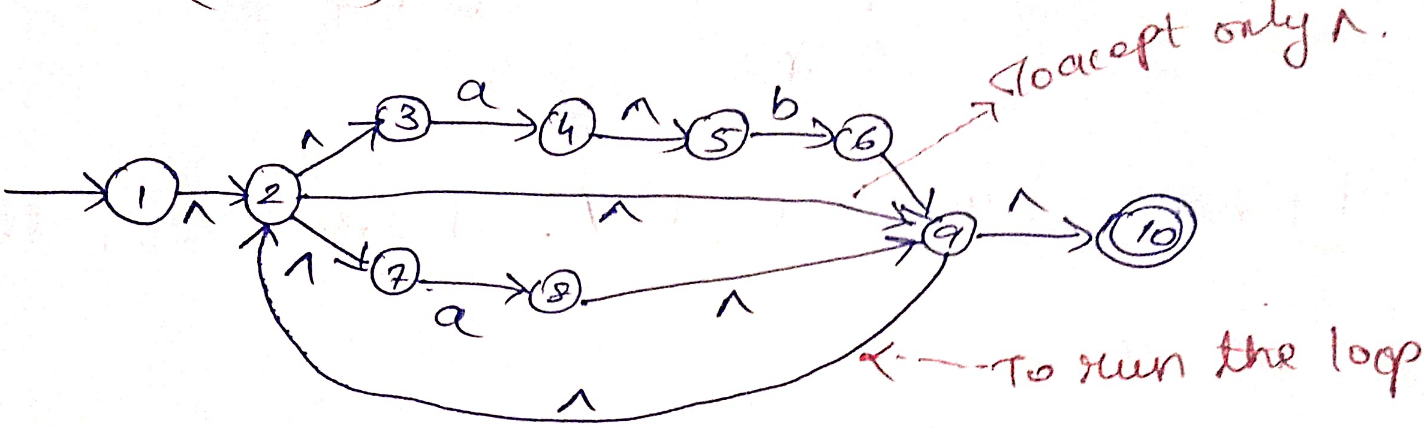
∴ $RE = a^*$

For *, to accept only ∧.

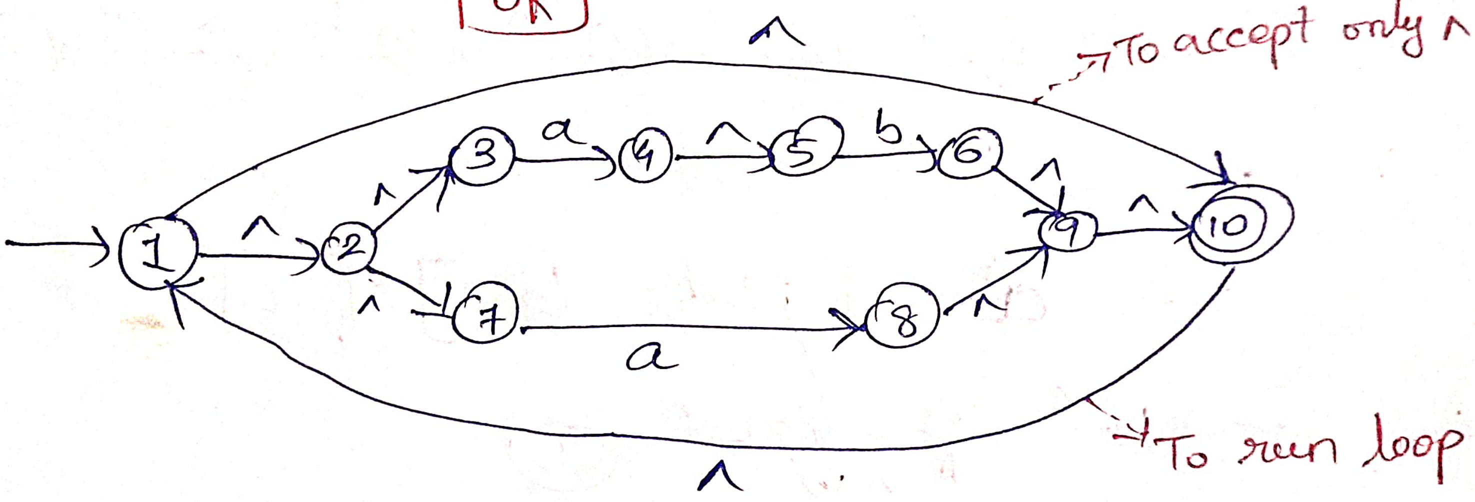
For making loop of a.

Eg Using Kleen's theorem, Draw ~~NFA~~, NFA-Λ for following:

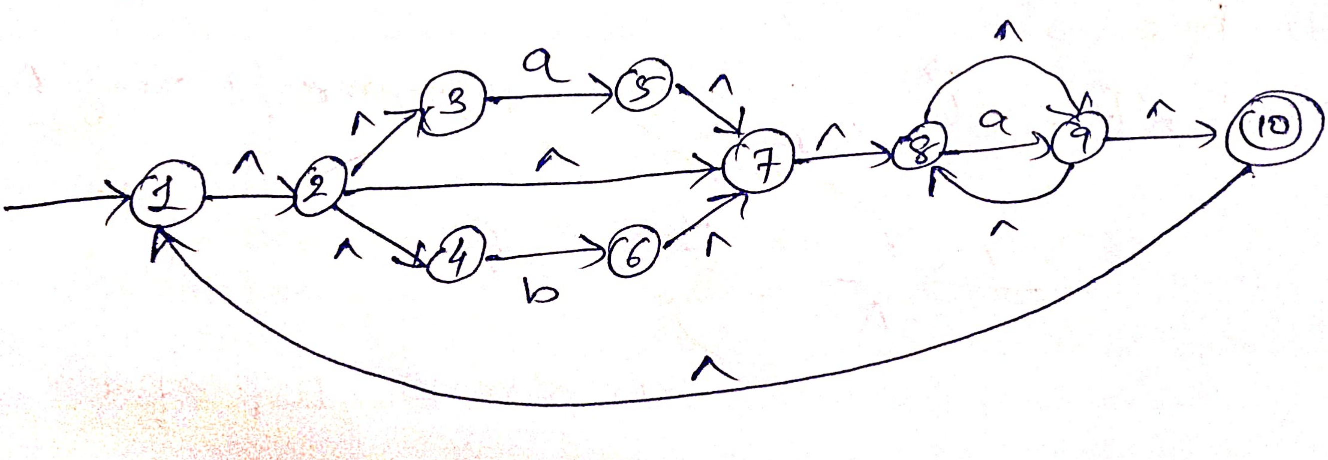
(1) (ab/a)*



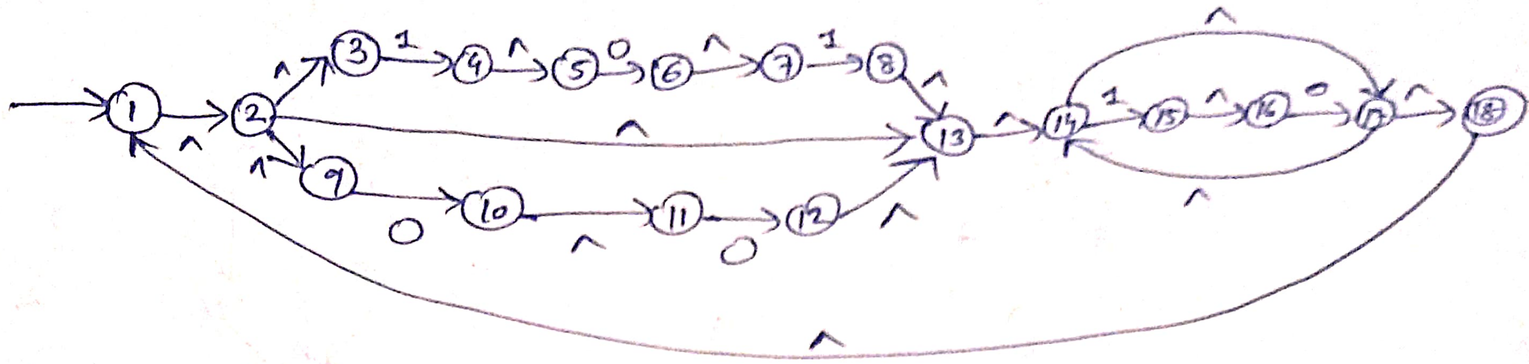
OR



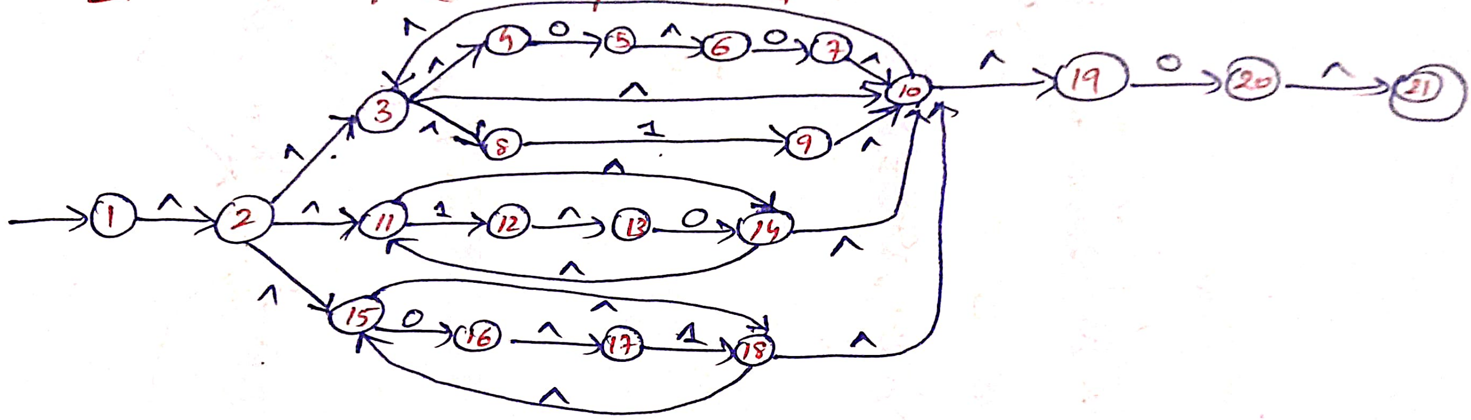
(2) [(a/b).a*]*



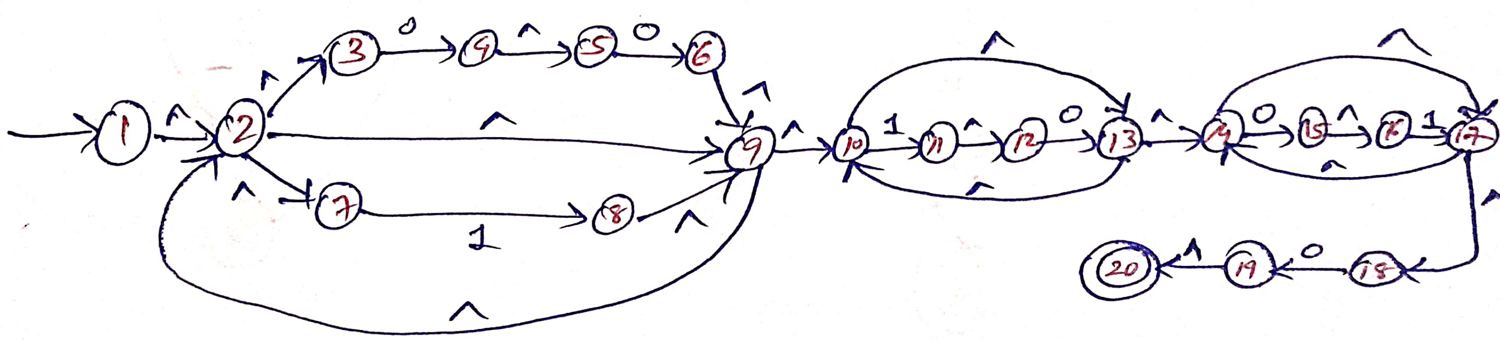
(3) $[(010/00)^*(10)^*]^*$



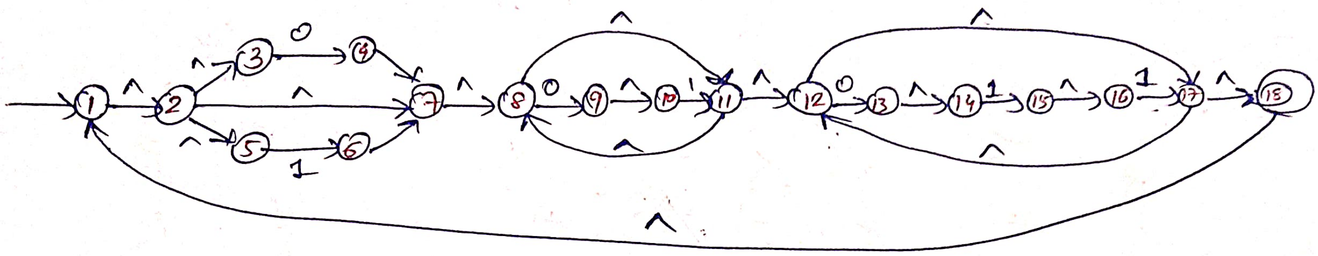
(4) $[(00/1)^* / (10)^* / (01)^*]0$



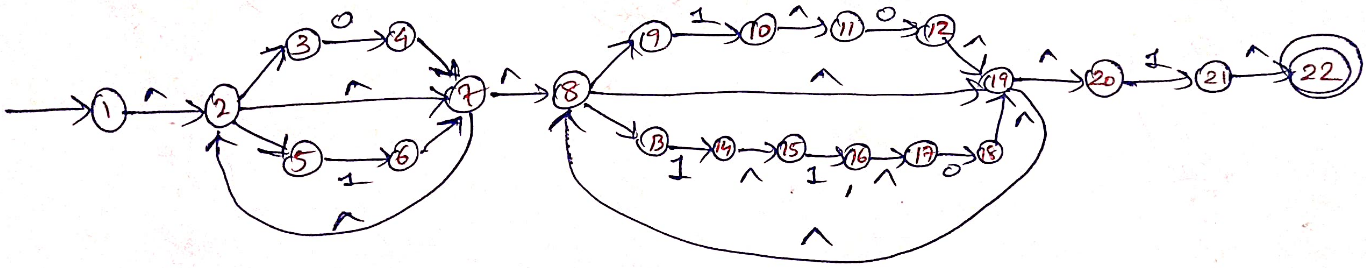
(5) $(00/1)^*(10)^*(01)^*0$



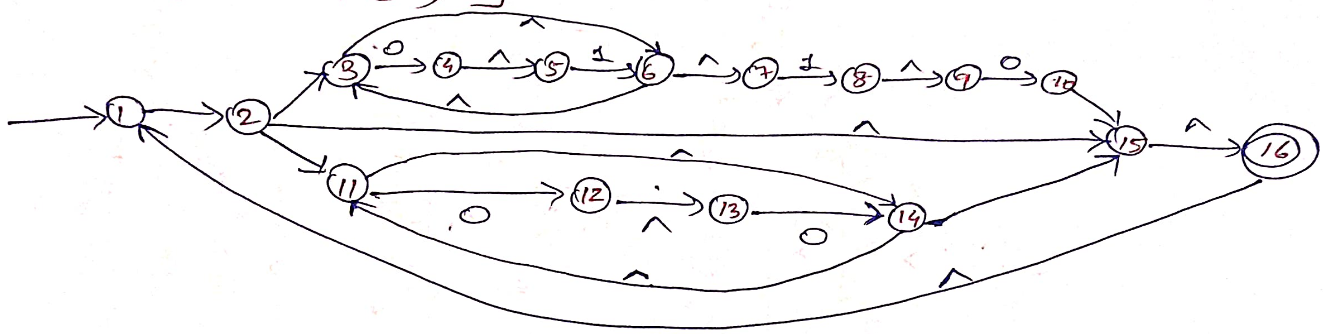
Eg [(0/1) (01)* (011)*]*



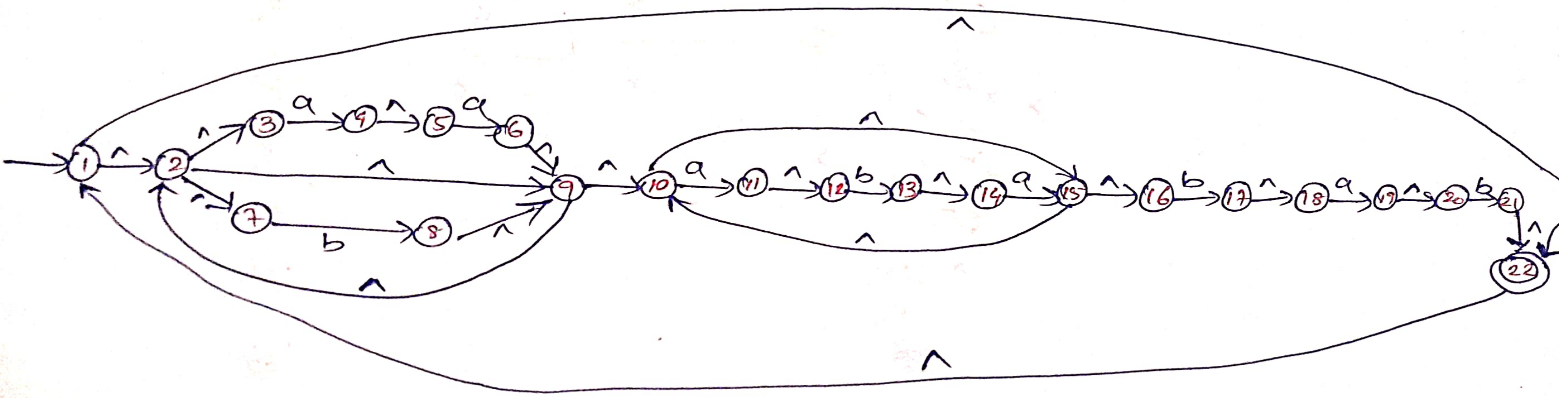
Eg (0/1)* (10/110)* . 1



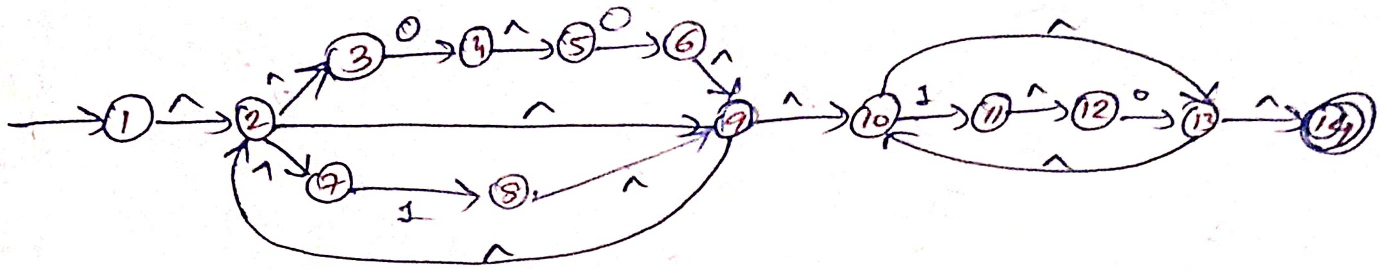
Eg $[(01)^* 10 / (00)^*]^*$



Eg $[(aa/b)^* (aba)^* bab]^*$



Eg $(00/1)^*(10)^*$



Eg $[\{10\} \cup \{00\}^* \{11\}^*]^*$

Sol^n $[\{10\} / \{00\}^* \{11\}^*]^*$